2 July 1979

MILITARY SPECIFICATION

GRIP ASSEMBLY, CYCLIC CONTROLLER, HELICOPTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cyclic controller grip assembly for use in helicopters.

1.2 Classification. The grip assembly shall be of the following styles, as specified on the procurement document.

Style 1	Grip Assembly, Model 76 Cyclic Controller Drawing 76-7477-1 (Pilot/Co-Pilot)
Style 2	Grip Assembly, Model 76 Cyclic Controller Drawing 76-7477-2 (Co-Pilot/Gunner)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, except those for which a particular issue is prescribed, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Cdr, US Army Aviation Research and Development Command, P.O. Box 209, St. Louis, MO 63166) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1680

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

MIL-P-116	Preservation, Methods of
DOD-D-1000	Drawing, Engineering and Associated List
MIL-E-6051	Electromagnetic Compatibility Requirements, System
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-M-18012	Markings for Aircrew Station Displays, Design and Configuration of
MIL-L-46159	Lacquer, Acrylic, Low Reflective
MIL-W-81044	Wire, Electric, Polyalkene - Insulated Tin - Coated Copper, NOrmal weight 150°c

STANDARDS

MILITARY

MIL-STD-1OO	Engieering Drawing Practices
MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking for US Military Property
MIL-sTD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System
MIL-STD-810C	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

DRAWINGS

ARMY

76-7477-1, -2	Grip Assembly, Cyclic Controller, Helicopter

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 First Article. Unless otherwise specified (see 6.2), this specification requires five complete aircraft controller grip assemblies for first article inspection and testing (see 4.4). First articles may be

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.

either preproduction samples or initial production models which conform to all the requirements of this specification. In either case, first articles shall be identical with the production. items in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish grip assemblies in accordance with this specification.

3.2 Quality Conformance. This specification required verification of quality conformance by sample production units (see 4.5).

3.3 Design and Construction.

3.3.1 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification and compatible as an assembly,

3.3.2 Basic Grip Design. The basic grip shall be an assembly comprised of a minimum of three molded plastic sections: a main body section, a left hand module, and a right hand module. The bodyand head modules shall have an internal configuration and receese sutable for retention of the switches in their respective positions and for routing of the wiring through. the mounting provisions of the grip assembly. The head modules shall be removable for access to the switches and wiring connections contained therein. A means shall also be provided for removing and replacing switches in the main body section. A means of mounting shall be provided at the base of the body, as shown on drawing 76-7477-1, -2. The fitting shall be suitable for insertion into or onto the end of the control stick. Color of the basic assembly shall he low-reflective black. MIL-L-46159 lacquer is prescribed for use as the coating for the basic assembly.

3.3.3 Electric Switches.

3.3.3.1 Electric switches incorporated into the grip assembly shall be of three general types, as follows:

a. Push button

b. Toggle, two options:

(1) Four positions on, center off
(2) Two positions on, center off

c. Trigger

3.3.3.2 The function and positioning of the switches shall be as illustrated in Figure 1, or as specified by the procuring activity. For helicopters, function and positioning of certain of the switches are prescribed as follows:

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T00:01Z
Check the source to verify that this is the currentversion before use.

a. Communications (intercom and radio) control only by fore-finger trigger switch.

b. Trim and stabilization by push button switches an left side (thumb rest position) of grip body.

c. Panel lights switching (between normal and night vision goggle levels) by push button on upper right switch head.

d. Critical function, such as weapon firings by push button under thumb guard in center of switch head.

3.3.3.3 The configuration of switch contacts, normally open (NO) or normally closed (NC), shall be specified for each switch used in the grip.

3.3.3.4 To the maximum extent possible, switches used for the grip assembly shall be designs which have been qualified and listed on pertinent qualified products lists (QPL). Before a switch can be selected fur use in the grip assembly, it shall have satisfied all requirements of 3.4.1.2. All switches shall have actuation forces and switch actuation travel within the limits specified in 4.7.4.

3.3.4 Switch Guard. The switch guard shall be spring loaded to remain in the closed position when the guarded switch is not in use. It shall be designed to be opened vertically by the pilot's thumb when he is wearing an authorized gloves. The guard shall be free to rotate through 90° +10° -0° and shall maintain a spring loading of 2 ±1/2 ounces of force throughout its travel. For guard dimensions refer to drawing 76-7477-1, -2.

3.3.5 Assembly Configuration. The assembly configuration shall be as shown on drawing 76-7477-1, -2.

3.2.6 Electromagnetic Interference. The grip assembly shall be designed using MIL-STD-461A (Notice 4) as a guide to achieve system compatibility when integrated into the aircraft. For incorporation into the grip assembly all switches, regardless of prior qualification, shall demonstrate satis-factory performance in accordance with the requirements of 3.3.6.1 and 3.3.6.2.

3.3.6.1 Electromametic Compatibility, Radiated Emission. Measurements shall be made in accordance with MIL-STD-461A, (Notice 4) method RE 02. The following conditions shall apply:

a. Loads switched shall be dc resistive at 28 volts and rated current of switch or grip assembly wiring, whichever is less.

b. Switch shall be actuated with a nonconductive rod.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.

c. Switch shall be mounted in grip assembly; i.e., grip assembly shall be considered part of the switch housing.

d. Radiated emissions shall not exceed levels shown in Figures 13 and 18 of MIL-STD-461A (Notice 4).

3.3.6.2 Conducted Mission. Measurements shall be made in accordance with requirements of MIL-E-6051 (spikes). The following conditions shall apply:

a. Loads switched - dc resistive at 28 volts and rated current of switch or grip assembly wiring, whichever is less.

b. Switch shall be actuated with a non-conductive rod.

c. Switch shall be mounted in grip assembly; i.e., grip assembly shall be considered part of the switch housing.

d. Spikes (transients) shall not exceed +50 percent nor -150 percent of the dc line voltage.

3.3.7 Drawings. In addition to 76-7477-1, -2, drawings forming a part of this specification are contractor supplied engineering design drawings as approved by the contracting officer. Each supplier is responsible for preparing his own shop drawings as necessary. Where tolerance prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances with those prescribed by contracting officer approved drawings to insure correct fit, assembly, and operation of the grip. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Contractor prepared drawings which are required for submittal to the government shall be in accordance with DOD-D-1000 and shall include linear dimensions and tolerances in both US customary units and metric units. Screw threads shall be specified in US customary units only.

3.3.8 Screw Threads. Unless otherwise specified, screw threads shall conform with MIL-S-7742.

3.3.9 Electrical Wiring. All hookup, interwiring, or cabling shall use wire conforming with MIL-W-81044 or MIL-W-16878, Type B.

3.3.10 Soldering, Unless otherwise specified, soldering shall conform with MIL-STD-454, Requirement 5.

3.3.11 Labels. Labels shall be attached at each switch to identify its function. The labels shall be removable so that legends may be altered in the event that switch function is changed, but shall be emplaced in a manner to insure against inadvertent removal in normal use or in cleaning. Letters and numerals on the labels shall be in accordance with MIL-M-18012.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

3.3.12 Weight. Weight of the -1 assembly shall not exceed 1.06 pounds exclusive of wiring. Weight of the -2 assembly shall not exceed 1.00 pound exclusive of wiring.

3.3.13 Selection of Specifications and Standards. Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Reliability.

3.4.1 Switches.

3.4.1.1. All switches used in the grip assembly specified herein shall be designed, constructed, and tested in accordance with current military or contractor specifications/drawings. In either case, the controlling specifications/drawings and any modifications thereto shall be as approved by the contracting officer.

3.4.1.2 Switch designs used in the grip assembly shall have passed inspections and tests prescribed by their respective specifications/drawings. These inspections and tests shall have established/evaluated, as a minimum, the following:

a. Actuation forces and travel (see 4.7.4, 3.6.1).

b. Dielectric verification (see 4.7.3).

c. Life cycle verification (50,000 cycles mechanical, 25,000 cycles electrical). For the communication (trigger type) switch life cycle requirements are 150,000 cycles mechanical, 75,000 cycles, electrical.

d. Ability to operate within specification limits after exposure to the following environment conditions (MIL-STD-810C).

(1) Humidity, Method 507

(2) Fungus, Method 508

(3) Salt Fog, Method 509

(4) Dust (fine sand), Method 510

3.4.2 Switch Guard Life. The switch guard life shall be a minimum of 150,000 operating cycles (see 4.7.5).

3.5 Maintainability. The grip assembly is desinged to permit ease of maintenance at field level, including replacement of switches, wiring, and associated parts without damage and without the use of special tools and techniques.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

3.6 Performance.

3.6.1 Switch Operation. All switches shall be operable without undue hand repositioning for normal operation of switches by the thumb and forefinger. The wearing of authorized gloves shall not unduly reduce the facility of operation of any switch nor the switch guard. Actuating force and actuator throw distance for the switches shall be as prescribed in their respective specifications. There shall be no binding of actuation nor limitation of travel due to fit of the switches on/in the body of the grip, On-off positioning of the switch actuator shall result in positive make-break contact action.

3.6.2 Dielectric Withstanding Voltage. When switches are tested as specified in 4.7.3, they shall withstand the application of the specified voltages without acring, breakdown of insulation, or visible damage to the switch parts. There shall be no flashover or leakage in excess of 1 milliampere.

3.6.3 Operation at Altitude. When tested as specified in 4.7.7.3, the switches shall withstand the application of the specified voltages without arcing, breakdown of insulation, or visible damage to the switch parts. There shall be no flashover or leakage in excess of 1 milliampere,

3.6.4 High Temperature. When the assembly is tested as specified in 4.7.7.4, there shall be no plastic deformation, separation, or breakage of the molded parts comprising the basic grip body (see 3.3.2). At the maximum temperature specified in 4.7.7.4 all switches shall operate in conformance with their respective design requirements.

3.6.5 Low Temperature. When the assembly is tested as specified in 4.7.7.5, there shall be no separation, spalling or cracking of the molded parts comprising the basic grip body (see 3.3.2) nor displacement of any of the switches in their mounts. At the minimum temperature specified in 4.7.7.5, all switches shall operate in conformance with their respective design requirements.

3.6.6 Strength of Assembly. When the assembly is tested as specified in 4.7.6, there shall be no breakage. There shall be no loosening of components or damage to the basic assembly nor to any of the installed parts which results in failure to function in accordance with 4.7.2 and 4.7.3.

3.6.7 Vibration. When the assembly is tested as specified in 4.7.7.1, there shall be no separation of closed switch contacts or closure of open contacts having a duration exceeding 10 microseconds. There shall be no evidence of damage to any part of the assembly which results in failure to function in accordance with 4.7.2, 4.7.3 and 4.7.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

3.6.8 Shock. When the assembly is tested as specified in 4.7.7.2, there shall be no breakage. There shall be no loosening of components or damage to the basic assembly nor to any of the installed parts which results in failure to function in accordance with 4.7.2, 4.7.3 and 4.7.4.

3.7 Identification Marking.

3.7.1 Identification of Product. Unless otherwise specified by the applicable drawing or contract, the grip assembly shall be marked for identification in accordance with MIL-STD-130 (minor assembly).

3.7.2 Part Numbering of Interchangeable Parts. All items having the same part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern. Part numbers utilized shall conform to drawing 76-7477-1, -2.

3.8 Workmanship. The grip assembly shall be constructed and finished in a thoroughly professional manner. The grip shall be free from burrs, sharp corners, and overlapping edges on mating surfaces. Particular attention shall be given *to* neatness and thoroughness of soldering, wirings marking of parts and assemblies plating, painting, riveting, machine screw assembly, welding, and brazing. Fastening devices such as screws and pins shall not interfere with handling the unit in operational service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Candidate Switches.

4.1.1 Certification Tests. For candidate switches the contractor shall submit for government approval a test plan for accomplishment of the tests prescribed by 3.3.6.1, 3.3.6.2, and 3.4.1.2. The contractor is then responsible for certification of suitability of the candidate switches through test in accordance with the approved test plan. Test results shall be presented to the Government contracting officer for approval.

4.1.2 Subsequent Program. The testing program below is for the full assembly as described by drawings 76-7477-1, -2 incorporating, as pertinent, QPL listed switches and other switches certified as specified in 4.1.1.

4.2 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other government approved facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

4.2.1 Component and Material Inspection. The supplier is responsible for insuring that components and materials used are manufactured, inspected and tested in accordance with referenced specifications and standards.

4.2.2 Disassembly Inspection. The first article grips failing a performance test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and the referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.3 Classification of Inspection. Inspections shall be classified as follows:

a. First article

b. Quality of conformance

c. Inspection of preparation for delivery

4.4 First Article Inspection.

4.4.1 First article. The five test articles selected as specified in 3.1 shall be individually inspected as specified in 4.6, and then subjected to the tests as described in 4.7.2 thru 4.7.7.6 in the sequence specified in Table I.

4.4.2 First Article Acceptance. Assembly of hardware produced/procured for the end item under contract, and production/procurement of parts for the assembly shall not proceed, except as authorized by the procuring contracting officer, until first article tests per 4.4.1 have been successfully completed.

4.4.3 Test Report. The contractor shall prepare a first article test report in accordance with MIL-STD-831 (see 6.3). The test report shall be submitted to the procuring activity for approval (see 6.2f).

4.5 Quality Conformance Inspection. Sample grip assemblies from production lots shall be subjected to quality conformance inspections as prescribed in 4.6.1, 4.7.2 and 4.7.3.

4.5.1 Critical Defects. Defects exposed in 4.6.1, Examinations 1, 2 and 3 and in 4.7.2 and 4.7.3 are established as critical defects.

4.5.2 Major Defects. Defects exposed in 4.6.1, Examinations 101 and 107 are established as major defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

4.5.3 Samples and Inspection Criteria.

4.5.3.1 Critical Defect Inspection. Five percent of each production lot (lot size and sampling procedure to be established by the procuring contracting officer) shall be inspected for critical defects. If one test sample fails to pass one or more of the tests, the lot shall be rejected.

4.5.3.2 Major Defect Inspection. The sampling plan for major defect inspection shall be in accordance with MIL-STD-105D, Table II-B. Acceptable quality level shall be 4.0.

4.5.4 Correction Procedures.

4.5.4.1 Critical Defect Rejections. When a production lot has been rejected for failure to pass the critical defect inspection, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made to all affected items, testing shall resume, including test/retest of all corrected items.

4.5.4.2 Major Defect Rejections. When a production lot has been rejected for failure to pass the major defect inspection, individual tests may be continued pending the investigation of the cause for rejection. However, final acceptance of items on hand or later produced shall not be made until it is determined that reworked items, individually, meet all requirements of this specification and production controls have been adjusted to correct the cause(s) for rejection.

4.5.5 Test Data Reporting. Quality conformance inspection and test data shall be provided in accordance with instructions in the procurement document.

4.6 Inspection Procedure.

4.6.1 Examination. The grips shall be examined for the following defects:

Critical.

1. Holes in castings.

2. Improper functioning of switches (3.6.1), including operation with the gloved hand.

3. Component parts missing or not as specified (3.3.1, 3.3.4, 3.3.7).

T e x t

101. Dimensions not as specified (3.3.2, 3.3.7).

102. Mounting not as specified (3.3.2. 3.3.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

103. Weight not as specified (3.3.12).

104. Interface fit of component parts not as specified (3.3.2, 3.3.7).

105. Identification and other markings illegible, missing, or not as specified (3.3.11, 3.7.1, 3.7.2).

106. Treatment not as specified (3.3.2, 3.3.7)

107. Workmanship not as specified (3.8).

4.7 Tests.

4.7.1 Test Conditions.

4.7.1.1 Standard Atmosphere Conditions. Wherever pressure and temperature are not specified for a test condition, it is understood that the test is to be made at standard atmospheric pressure (approximately 29.92 inches Hg) and at normal room temperature (approximately 25°C or 77°F).

4.7.1.2 Test Fixture. or those tests requiring a fixture (such as strength, vibration, and shock) the grip assembly shall be mounted to the fixture by the normal mounting provisions of the grip assembly being tested as specified by the applicable drawing.

4.7.2 Operational Check. Each switch of the grip assembly shall be electrically connected to its rated voltage per basic switch rating. Each switch shall be operated at least 5 times to insure correct operation and secure mounting. A single actuation failure or evidence of improper switch orientation or insecure mounting shall be cause for rejection.

4.7.3 Dielectric Test. Each switch, when installed in the grip assemply shall be subjected to a 60-cycle frequency having a potential of 1060 volts root-mean-square (RMS). For first article testing, the length of time shall be 1 minute and for conformance testing, the lenth of time shall be 1 second. Test as follows:

a. Between all terminals and exposed or grounded metal parts.

b. Between all adjacent terminals of different poles, if any.

c. Between all open terminals and corresponding pole.

These tests shall be performed with the switch in the normal position, and shall then be repeated for all switch positions. Any evidence of arcing, flashover, breakdown of insulation, or current flow in excess of 1 milliamp shall be an indication of failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

4.7.4 Switch Force and Travel. Each switch in the grip assembly shall be tested to determine that actuating force and travel is in accordance with its applicable drawing.

4.7.5 Switch Guard Life Test. The Switch Guard shall be life tested per requirements of 3.4.2 by mechanically simulating the action of the pilot's gloved thumb flipping up the guard from the left and removing his thumb to the left, allowing the spring loaded guard to return to its normal, fully closed position as representing one cycle. Breakage or failure to operate as specified in 3.3.4 shall be indication of failure.

4.7.6 Strength Test. The grip assembly shall be mounted on a test fixture by the base mounting applicable to the assembly type specified, The following loads shall be applied over a small area at the reference point, as shown on the applicable grip assembly drawings. A fixture may be clamped around the grip at the reference point.

a. 300 pounds forward.

b. 300 pounds aft.

c. 150 pounds to the right.

d. 150 pounds to the left.

e. 100 pounds upward.

f. 100 pounds downward.

g. 80-inch-pounds torque.

There shall be no cracking, breaking or loosening of components resulting in malfunction of the grip assembly in accordance with 4.7.2 and 4.7.3.

4.7.7 Environmental Tests. The environmental tests of paragraph 4.7.7.1 through 4.7.7.6 shall be conducted in accordance with MIL-STD-810C as modified herein.

TEST	METHOD	PROCEDURE
Vibration	514	I, Category C, Table 514-2-III, Fig 514.2-3, Curve M, (4.7.7.1)
Shock	516	I, Fig 516.2-1, (a) & (c) (4.7.7.2)
Altitude		(4.7.7.3)
High Temperature	501	I, (4.7.7.4)

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.

TEST	METHOD	PROCEDURE
Low Temperature	502	I, (4.7.7.5)
Temperature Shock	503	I, (4.7.7.6)

4.7.7.1 Vibration. During vibration tests, there shall be no separation of closed switch contacts or closure of open contacts having a duration exceeding 10 microseconds. At the conclusion of the vibration tests, the grip assembly shall be tested as described in 4.6, 4.7.2, 4.7.3, and 4.7.4 for indication of failure. Mounting of the assembly shall simulate in-service mounting with respect to attachment and orientation.

4.7.7.2 Shock. Mounting of the test assembly shall simulate in-service mounting with respect to attachment. The peak shock value (A) shall be 20 g's. Three shocks shall be applied to the test item (see 3.6) in each direction, along each of the three mutually perpendicular axis (18 shocks). At the conclusion of the shock tests, the grip assembly shall be tested as described in 4.6, 4.7.2, 4.7.3, and 4.7.4 for indication of failure. Shocks shall be applied at standard atmospheric conditions as follows:

a. Vertically - three shocks in each direction.

b. Parallel to the major horizontal axis - three shocks in each direction.

c. Parallel to the minor horizontal axis - three shocks in each direction.

4.7.7.3 Altitude. While the grip assembly is at the lowest pressure condition-of 13.75 inches Hg, it shall be tested as specified in 4.7.3 at a reduced potential of 500 volts. The rate of pressure change during decompression shall be 1.5 inches of mercury per minute. Any evidence of arcing, flashover, breakdown of insulation, or current flow in excess of 1 milliampere shall be indication of failure.

4.7.7.4 High Temperature. While the grip assembly is at the operating temperature of 71°C (160°F), it shall be tested as described in 4.7.2. The grip assembly shall be immediately subjected to the strength test (4.7.6) after removal from the 71°C (160°F) test chamber. The strength test shall be completed within 10 minutes after removal from the test chamber. Following the strength test, the assembly shall be subjected to the tests described in 4.6, 4.7.2, 4.7.3, and 4.7.4.

4.7.7.5 Low Temperature. While the grip assembly is at the operating temperature of minus 54°C (minus 65°F), it shall be tested as described in

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.

4.7.2. The grip assembly shall be immediately subjected to the strength test (4.7.6) after removal from the minus 54°C (minus 65°F) test chamber. The strength test shall be completed within 10 minutes after removal from the test chamber. Following the strength test, the assembly shall be subjected to the tests described in 4.6, 4.7.2, 4.7.3, and 4.7.4.

4.7.7.5 Temperature Shock. For steps 1, 4 & 5 of the temperature shock test, the high temperature used shall be 88°C (190°F). At the conclusion of the temperature shock test, the grip assembly shall meet the strength test described in 4.7.6 and tests described in 4.6 and 4.7.2.

4.8 Inspection of Preparation for Delivery. Preservation and packaging, packing, and marking of the grip assembly shall be inspected to determine conformance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation and Packaging. Preservation shall be Level A or E, when and as specified (see 6.2).

5.1.1 Level A. Each grip assembly shall be preserved and packaged in accordance with MIL-P-116, Method IAS (without the use of a contact preservation) in a unit container conforming to W5s and W5c of PPP-B-636, The packaged grip assemblies shall be cushioned to limit movement within the container.

5.1.2 Level B. Each grip assembly shall be preserved as for Level A (without the use of a contact preservative) and placed in a unit container conforming to domestic class of PPP-B-636, cushioned to limit movement within the container.

5.2 Packing. Packing shall be Level A or E, when and as specified (see 6.2).

5.2.1 Level A. Grip assemblies packaged as specified in 5.1 shall be packed in overseas-type shipping containers conforming to PPP-B-601. Insofar as practicable, exterior shipping containers shall be of uniform shape and size, of mininmm cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each container shall not exceed 200 pounds.

5.2.2 Level B. Grip assemblies packaged as specified in 5.1 shall be packed in containers conforming to V3C or V3S of PPP-B-636. Containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior containers shall be of uniform shape and size and shall contain identical quantities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

5.3 Marking of Shipments. Levels A and B packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and as specified in the contract or order (see 6.2).

5.4 Commercial Packaging. When specified (see 6.2) packaging, packing, and marking, as applicable, shall be acceptable if they meet the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended Use. The controller grip assembly covered by this specification is intended for mounting on the cyclic control stick of a helicopter to provide a handgrip for the pilot/copilot gunner and contains the switches necessary to operate certain control functions and weapons systems.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Style of grip assembly required (see 1.2).

Level of packaging and packing required (see 5.1 and 5.2), or commercial packaging (see 5.4).

d. Whether special marking for shipment is required (see 5.3).

e. Whether First Articles are required (see 3.1 and 4.4.1).

f. Dates for delivery of test plans, data, and reports, as required.

g. Contractor prepared drawings.

6.3 Test Reports. The test reports (see 4.4.3, 4.5.5) shall contain data sufficient to establish. all performance criteria under test, test parameters, ambient and controlled test environments, test methods and procedures, test item identification, date of test, location of test facilities, identification of test personnel, quantitative data, and defects in accordance with 4.5.1 and 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

TABLE I. ORDER OF TESTING

TESTS		FIRST ARTICLE TEST SEQUENCE SAMPLE NUMBERS				
	PARA	1	2	3	4	5
Examination of Product	4.6	1 (7)	1 (7)	1 (7)	1 (7)	1 (7)
Operational Check	4.7.2	2 (8)	2 (8)	2 (5*)(8)	2 (5*)(8)	2 (8)
Dielectric	4.7.3	3 (9)	3 (9)	3 (9)	3 (9)	3
Switch Travel and Actuating Force	4.7.4	4 (10)	4 (10)	4 (10)	4 (10)	
Switch Guard Endurance	4.7.5	5				4
Strength	4.7.6			6	6	6
Vibration	4.7.7.1		6			
Shock	4.7.7.2	6				
Altitude	4.7.7.3		5			
High Temperature	4.7.7.4			5*		
Low Temperature	4.7.7.5				5*	
Temperature Shock	4.7.7.6					5

NOTE: NUMBERS IN PARENTHESIS INDICATE REPEAT TEST TO BE CONDUCTED IN SPECIFIED SEQUENCE.

*TEST PER 4.7.2 (TEST SEQUENCE 5) TO BE CONDUCTED AFTER SPECIFIED TEMPERATURE HAS BEEN MAINTAINED FOR ONE HOUR.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.

6.4 International Standardization. Certain provisions of this specification are the subject of international standardization agreements ASCC AIR STD 10/12, Operation of All Controls and Switches at Aircrew Stations in Fixed Wing and Rotary Wing Aircraft and ASCC AXR STD 10/22, Services operable from Stick Grip in Both Fixed Wing and Rotary Wing Aircraft. When amendment, revision, or cancellation of this specification is proposed, which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Reclaimed Materiels. The use of reclaimed materials shall be encouraged to the maximum extent possible.

CUSTODIANS:

Army - AV

Air Force - 11

Navy - AS

PREPARING ACTIVITY:

Army - AV

Project NO. 1680-0450

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:01Z
Check the source to verify that this is the current version before use.

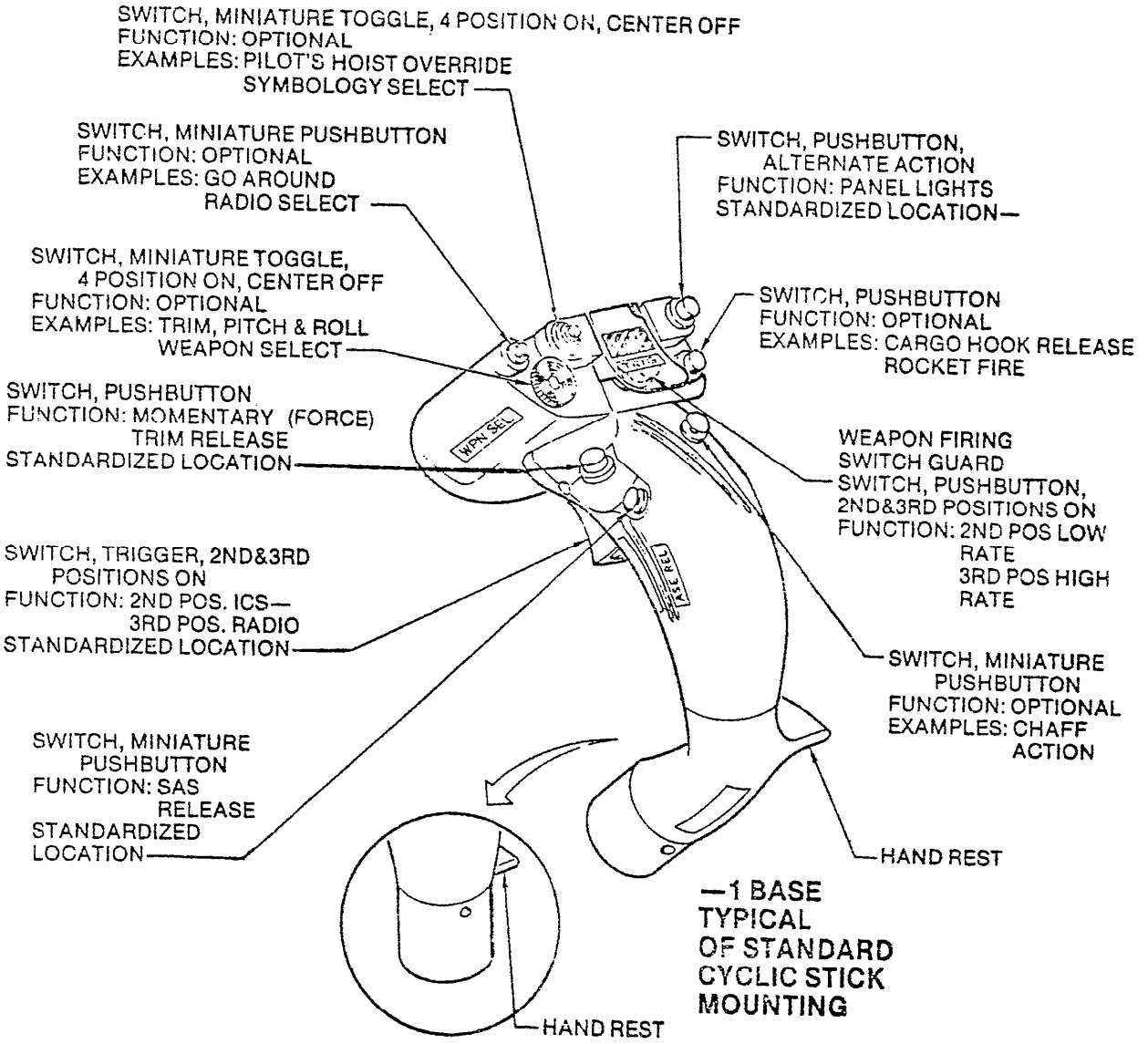


FIGURE I

Illustration of Controller Assembly Showing
Standard Switch Locations and Possible Options

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-G-58103 Basic

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:01Z
Check the source to verify that this is the current version before use.